

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 221 — Past Questions & Materials
by Renance

Course: MTH 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — MTH 221 questions for revision

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1. Solve the IVP $\square\square = \square\square - \square \sec(\square)(1 + \square^2)$, $\square(0) = 0$

−3

1 −3

2. Show that $\square(\square) = \square^2$ is a solution to $4\square^2\square'' + 12\square\square' + 3\square = 0$, $\square(4) = 8$, $\square'(4) = 64$.

3. Derive the formula used to find the integrating factor for the solution of a linear first order differential equation.

4. Find the solution to the IVP $2\square' - y = 4 \sin(3\square)$, $\square(0) = \square 0$. [7 mks]

5. Find the solution and interval of validity to the following IVP

$2\square\square$

$- 2\square - (2 - \ln(\square^2 + 1))\square' = 0$, $\square(5) = 0$.

$\square^2 - 1$

□

6. Find the solution and interval of validity to the following IVP $\square' + \square - \sqrt{\square} = 0$, $\square(1) = 0$.

7. A 1500 gallon tank initially contains 600 gallons of water with 5 lbs of salt dissolved in it. Water enters the tank at a rate of 9 gal/hr and the water entering the tank has a salt

concentration of $5(1 + \cos(\square))$ lbs/gal. If a well-mixed solution leaves the tank at a rate of 6 gal/hr, how much salt is in the tank when it overflows?

8. A 50 kg object is shot from a cannon straight up with an initial velocity of 10m/s off a bridge that is 100 meters above the ground. If air resistance is given by $5\square^2$ determine the velocity of the mass at any time t.

9. Write down the following 4th order differential equation as a system of first order, linear differential equations and convert the system into matrix form. $\square(4) + 3\square'' - \sin(\square)\square' + 8\square = \square^2$, $\square(0) = 1$, $\square'(0) = 2$, $\square''(0) = 3$, $\square'''(0) = 4$

10. Convert the following differential equation into a system, solve this system and use this solution to get the solution to the original differential equation. $2\square'' + 5\square' - 3\square = 0$, $\square(0) = -4$, $\square'(0) = 9$

11. State a theorem which gives sufficient conditions gives sufficient conditions for existence and uniqueness of solutions of solutions of initial value problems

12. Apply the theorem in 11. above to the IVP below; determine if it has a unique solution on an open

interval that contains $\square 0 = 0$. Find the solution and determine the largest open interval on which it is unique

1

$\square' = 3x(\square - 1)^3$, $\square 0 = 9$

13. Find infinitely many solutions of the IVP above, all defined on $(-\infty, \infty)$.